Superpatterns: certified bounds, structured classes, and a layered obstruction to Alon's conjecture

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Abstract

We study the minimum length $\mathrm{sp}(k)$ of a permutation containing every pattern of length k, and the length at which a random permutation does. A stable value-slot encoding, with a corrected finite-width reduction and an outward interval certificate, proves $\mathrm{sp}(k) > 1.0073k^2/e^2$ for all sufficiently large k. Explicit witnesses give $\mathrm{sp}(7) \le 23$ and $\mathrm{sp}(8) \le 30$. A directed MPFR certificate supports a typical-target containment bound at every coefficient above 0.4649. At Alon's coefficient $1/4$ we prove simultaneous containment for monotone inflations with blocks of length at least $K\sqrt{\log k}$ and, under a revisit-separation hypothesis, for tilted grids. Our main new observation concerns layered patterns. Replacing a block 12 by 21 in a direct sum never increases the probability of containment; hence the number L_{21} of direct-summed copies of 21 in a random permutation is stochastically dominated by $\lfloor \mathrm{LIS}/2 \rfloor$, and $c_{21} = \lim L_{21}/\sqrt{n} \le 1$. Paired simulations for n from 10^3 to 10^8 give $\mathbb{E}\lfloor \mathrm{LIS}/2 \rfloor - \mathbb{E}L_{21} = (0.00113 \pm 0.00002)\sqrt{n}$, that is $c_{21} \approx 0.99887$. If $c_{21} < 1$, Alon's conjecture fails at the sharp coefficient $1/4$; we reduce the strict inequality to an explicit lemma but do not prove it. Claims of general universality at Ck^2 made in an earlier draft are withdrawn.

1 Scope and results

A permutation $\sigma \in S_n$ contains $\pi \in S_k$ if some subsequence of σ has relative order π. It is a *k-superpattern* if it contains every member of S_k; write $\mathrm{sp}(k)$ for the minimum possible length and $\mathrm{pat}_k(\sigma)$ for the number of distinct contained k-patterns.

The deterministic question and the random universality question have different constants. Chroman, Kwan and Singhal prove the deterministic lower bound with coefficient $1.000076/e^2$ [1]; the general construction of Engen and Vatter has length $\lceil (k^2+1)/2 \rceil$ [2]. Alon's conjecture asks whether a uniform permutation of length $\lceil (1/4+\varepsilon)k^2 \rceil$ simultaneously contains all of S_k, with probability tending to one, for every fixed $\varepsilon > 0$. He and Kwan establish this with length $2000k^2 \log\log k$ [3]. A uniform bound tending to zero on the failure probability of each individual target does not by itself imply simultaneous containment.

For individual and typical targets, the relevant recent comparison is the online embedding work of Altschuler, Dubroff and Tikhomirov [4, Thms 1.3, 1.5, 1.7]. Their concentration results imply offline containment at coefficients $0.50568+\varepsilon$ uniformly over individual targets, and $0.49967+\varepsilon$ for a uniform target. This deduction standardizes a sufficiently long initial segment of the iid stream; a fixed quadratic margin absorbs their fluctuations. These results supersede the older 0.757 and 0.527 benchmarks in our workstream logs.

Table 1: Selected current results. The original theorem numbers are retained to match the experiment archive. "Certificate" here is numerical unless Lean is explicitly named.

Result	Comparison	Current statement and verification
Deterministic lower bound, C'	$1.000076/e^2$ [1]	$1.0073/e^2$; gain $0.007224/e^2$; analytic proof and Decimal intervals
Stable encoding A	Same baseline	Explicit finite inequality; Lean. Its Lean numerical certificate uses $\lambda = 1.0003$
Short witnesses, Thm 2	Lengths $25, 33$ [2]	Lengths $23, 30$; exhaustive check and Lean with `native_decide` trust
Typical target, Thm 17	$0.49967 + \varepsilon$ [4]	Every $C > 0.4649$; reduction and directed MPFR supersolution
Independent-strip cost, Thm 20	Earlier Monte Carlo conjecture	$n \le \mathbb{E}K_n \le n + \sqrt{2n} + 1/2$; analytic proof
Simultaneous tilted grids	Earlier corner rule: $\pi/8$	Every $1/4 + \varepsilon$, with $\min(r, h) \ge \log^3 k$; repaired reduction
Simultaneous monotone inflations, Thm 22	Earlier defect-free box argument required $\sqrt{k}\log k$ blocks	$K\sqrt{\log k}$ blocks, arbitrary skeleton; shared squares and [5]
Repeated-21 frontier, Props. 23–24	Original per-level implementation	Exact $O(n \log n)$ algorithm, linear space; every host through length 8 checked
Block flip, Thm 26	Skew shapes [6]; 1324 vs 1234 [7]	Flipping $12 \to 21$ in a direct sum never raises containment; exact fibre formula; all of S_n, $n \le 8$, checked
Repeated 21, Cor. 27	Earlier here: $c_{21} \le 1.140$	$c_{21} \le 1$ and $L_{21} \le_{st} \lfloor \mathrm{LIS}/2 \rfloor$; proof
Repeated 21, numerics	Alon needs $c_{21} \ge 1$	$c_{21} \approx 0.99887$ from paired samples, $n \le 10^8$; not a proof

The certificate files and exact quantifiers are indexed in the repository's `memory/RESULTS.md`. A dedicated note, *An exact frontier for repeated 21*, supplies a self-contained proof of Propositions 23–24 with a prior-art table; its PDF and Markdown are in `output/pdf/`. The block-flip inequality of §6.2 is the size-two transversal case of a theorem on fillings of skew shapes [6] and is not claimed as new; its asymptotic consequence for random permutations and its relation to Alon's conjecture appear to be. The shared-square corollary and frontier algorithm are likewise not asserted to be novel. Earlier versions overstated several conclusions, and a draft of September 2026 announced proofs of universality at Ck^2 and at the sharp constant that do not hold; the affected statements are identified in §7.

2 The lower bound

Throughout, k is odd (for even k, apply the odd result at $k-1$ and use $\mathrm{sp}(k) \ge \mathrm{sp}(k-1)$). A k-subset $T = \{t_1 < \cdots < t_k\} \subseteq [n]$ corresponds bijectively to the *gap vector* $a = (a_0, \dots, a_k)$, $a_0 = t_1$, $a_j = t_{j+1} - t_j$ $(1 \le j < k)$, $a_k = n + 1 - t_k$: a composition of $n+1$ into $k+1$ positive parts. The *width* of index i $(2 \le i \le k-1)$ is $b_i = t_{i+1} - t_{i-1} = a_{i-1} + a_i$.

2.1 Stable encodings

The starting point is the encoding idea of [1]: a pattern π occurring at T can be recorded by giving the positions t_j for $j \notin I$ and the *values* $\pi(i)$ for $i \in I$, which is cheaper than giving t_i whenever the window (t_{i-1}, t_{i+1}) has more than k integer points. CKS fix a threshold and a set of exactly ck indices and pay a large-deviation estimate for subsets with few wide windows. The following lemma removes the threshold and the case distinction.

Let $\mathcal{I}$ be a map $T \mapsto I(T) \subseteq \{2, \dots, k-1\}$ such that $I(T)$ contains no two consecutive integers and is *stable*: whenever T' agrees with T outside $I(T)$ (i.e. $t'_j = t_j$ for all $j \notin I(T)$), we have $I(T') = I(T)$.

Lemma 1. For every stable $\mathcal{I}$ and every $\sigma \in S_n$,

$$\mathrm{pat}_k(\sigma) \;\le\; \sum_{T \in \binom{[n]}{k}} \frac{k!}{(k - |I(T)|)!} \prod_{i \in I(T)} \frac{1}{b_i(T) - 1}.$$

Proof. For each π contained in σ fix one occurrence $T = T(\pi)$ and set $I = I(T)$, $\varphi(\pi) = (I, (t_j)_{j \notin I}, (\pi(i))_{i \in I})$. φ is injective: from σ and $(t_j)_{j \notin I}$ we know the relative order of the values $\sigma(t_j)$, $j \notin I$; from $(\pi(i))_{i \in I}$ we know which $k - |I|$ ranks in $[k]$ are taken by the indices $j \notin I$; together these determine π. Group the codes by $\psi = (I, (t_j)_{j \notin I})$; each ψ admits at most $k!/(k-|I|)!$ third components. Let $\mathrm{ext}(\psi)$ be the set of $T' \in \binom{[n]}{k}$ agreeing with ψ off I. Since I has no two consecutive elements, t'_i ranges independently over the open interval (t_{i-1}, t_{i+1}) for $i \in I$, so $|\mathrm{ext}(\psi)| = \prod_{i \in I}(b_i - 1)$, where b_i is determined by ψ. By stability, every $T' \in \mathrm{ext}(\psi)$ has $I(T') = I$, so T' determines ψ and the sets $\mathrm{ext}(\psi)$ are pairwise disjoint. Hence

$$\begin{aligned} \sum_{\psi} \frac{k!}{(k-|I|)!} &= \sum_{\psi} \sum_{T' \in \mathrm{ext}(\psi)} \frac{k!}{(k-|I|)!\,|\mathrm{ext}(\psi)|} \\ &\le \sum_{T'} \frac{k!}{(k - |I(T')|)!} \prod_{i \in I(T')} \frac{1}{b_i(T') - 1}. \end{aligned}$$ ∎

2.2 The even rule and the tilt

Take $I(T) = \{i \text{ even} : b_i - 1 > k\}$. It is non-adjacent, and stable because it depends only on the odd-indexed positions $t_{i\pm1}$, none of which lie in $I(T)$. Using $k!/(k-|I|)! \le k^{|I|}$, Lemma 1 gives

$$\mathrm{pat}_k(\sigma) \le \sum_T W(T), \qquad W(T) = \prod_{i \text{ even}} f(b_i), \quad f(b) = \min\Big(1, \frac{k}{b-1}\Big). \tag{1}$$

In gap coordinates, $W(a) = \prod_{j=1}^{(k-1)/2} f(a_{2j-1} + a_{2j})$: a product over *disjoint* pairs of gaps. For any $x \in (0,1)$, since $\sum a = n+1$ on every term,

$$\begin{aligned} \sum_T W(T) &= x^{-(n+1)} \sum_{a:\, \sum a = n+1} x^{\sum a} W(a) \;\le\; x^{-(n+1)} \sum_{a \in \mathbb{Z}_{\ge 1}^{k+1}} x^{\sum a} W(a) \\ &= x^{-(n+1)} \Big(\frac{x}{1-x}\Big)^2 \prod_{j=1}^{(k-1)/2} \sum_{a,a' \ge 1} x^{a+a'} f(a+a'). \end{aligned} \tag{2}$$

(The free gaps a_0, a_k contribute $\sum_{a \ge 1} x^a = x/(1-x)$ each.)

Lemma 2. For $x \in (0,1)$ and $k \geq 1$, $\displaystyle\sum_{a,a' \geq 1} x^{a+a'} f(a+a') = \Big(\frac{x}{1-x}\Big)^2 (1-x^k)$.

Proof. Writing $s = a + a'$, there are $s-1$ pairs with sum s. Split at $s = k+1$:

$$\sum_{s=2}^{k+1} (s-1)x^s + \sum_{s \geq k+2} (s-1)x^s \frac{k}{s-1} = \Big[\sum_{s \geq 2} (s-1)x^s - \sum_{s \geq k+2} (s-1)x^s\Big] + \frac{k\,x^{k+2}}{1-x}.$$

Now $\sum_{s\geq 2}(s-1)x^s = x^2/(1-x)^2$ and $\sum_{s\geq k+2}(s-1)x^s = x^{k+2}[(k+1)/(1-x) + x/(1-x)^2]$. Collecting terms over the common denominator $(1-x)^2$:

$$\frac{x^2 - x^{k+2}\big((k+1)(1-x)+x\big) + kx^{k+2}(1-x)}{(1-x)^2} = \frac{x^2 - x^{k+2}}{(1-x)^2} = \Big(\frac{x}{1-x}\Big)^2 (1-x^k).$$

Combining (1), (2) and Lemma 2:

Theorem A. For every $\sigma \in S_n$, every odd $k \geq 3$ and every $x \in (0,1)$,

$$\mathrm{pat}_k(\sigma) \;\leq\; x^{-(n+1)} \Big(\frac{x}{1-x}\Big)^{k+1} (1-x^k)^{(k-1)/2}. \tag{A}$$

Note that dropping the factor $(1-x^k)^{(k-1)/2}$ and optimizing x recovers exactly the trivial bound $\binom{n}{k}$ up to lower-order terms; the whole gain is in that factor.

2.3 Asymptotics

Put $n = \lambda k^2/e^2$ and $x = e^{-\theta/k}$ with $\theta > 0$ fixed. Then $-(n+1)\log x = (n+1)\theta/k$, $\log \frac{x}{1-x} = \log \frac{k}{\theta} - \frac{\theta}{2k} + O(k^{-2})$, and $\log k! = k \log k - k + O(\log k)$; hence

$$\frac{1}{k} \log \frac{\mathrm{RHS(A)}}{k!} = \underbrace{\frac{\lambda\theta}{e^2} - \log\theta + 1 + \tfrac{1}{2}\log(1-e^{-\theta})}_{=:g(\theta,\lambda)} + O\Big(\frac{\log k}{k}\Big). \tag{A$'$}$$

$g(\cdot,\lambda)$ is strictly convex on $(0,\infty)$ (indeed $g_{\theta\theta} = \theta^{-2} - [8\sinh^2(\theta/2)]^{-1} > 0$), with a unique minimizer $\theta^*(\lambda)$ solving $\lambda/e^2 = 1/\theta - \frac{1}{2}e^{-\theta}/(1-e^{-\theta})$, and $\min_\theta g(\theta,\lambda)$ is continuous and strictly increasing in λ. Numerically (mpmath, 30 digits, two independent implementations):

$$\theta^*(1) = 7.37192\ldots, \qquad g(\theta^*(1),1) = -3.1174 \times 10^{-4},$$
$$\lambda_A \approx 1.00031251, \qquad \theta^*(\lambda_A) = 7.36959\ldots$$

Here λ_A is defined by $\min_\theta g(\theta,\lambda_A) = 0$. Hence for every fixed $\lambda < \lambda_A$ and all large k, every $\sigma \in S_n$ with $n \leq \lambda k^2/e^2$ satisfies $\mathrm{pat}_k(\sigma) \leq e^{-ck} k!$ for some $c = c(\lambda) > 0$; in particular σ is not a k-superpattern.

Certified corollary of A. $\mathrm{sp}(k) \geq (1.0003 - o(1))\, k^2/e^2$. The stronger optimized value above is numerical; the Lean certificate establishes $g(7.37, 1.0003) < 0$. The asymptotic passage itself is the argument just given, not a separately formalized Lean corollary.

For a fully explicit finite check, at $k = 100001$, $\lambda = 1.00031$, $\theta = \theta^*(\lambda)$, the exact right-hand side of (A) gives $\frac{1}{k}\log(\mathrm{RHS}/k!) = -1.09 \times 10^{-5} < 0$.

Comparison with [1]. Their encoding specifies values for exactly ck even indices ($c = 0.00075$) whose widths exceed dn/k ($d = 8.180$), and needs their Lemma 2.1 (a Chernoff bound showing that all but $O(\lambda^k)\binom{n}{k}$ subsets have that many wide windows); this yields $\log \lambda_{\mathrm{CKS}} \approx c \log(d/e^2) \approx 7.6 \times 10^{-5}$. In Theorem A the right-hand side of Lemma 1 is an *expectation of a product*, so after the exponential tilt every wide window contributes its full saving $\min(1, \theta/B)$, $B \sim \Gamma(2,1)$ in the limit, with $\mathbb{E}\min(1,\theta/B) = 1 - e^{-\theta}$; no case split is needed. The tilt itself costs only $\tau - 1 - \log\tau \approx 2.7 \times 10^{-6}$ where $\tau = \theta\lambda/e^2 \approx 0.9977$.

CKS discuss refinements using additional indices and sharper large-deviation estimates. The present formulation packages the even-index savings as a product expectation. The improvement is within that encoding mechanism; priority assessment remains provisional.

2.4 The value-slot refinement

Lemma 1 charges index $i \in I$ the factor $1/(b_i - 1)$ because t_i has $b_i - 1$ possible positions, each of which could in principle give a different pattern. But the pattern only depends on where $\sigma(t_i)$ falls among the other chosen values, so the number of distinct patterns obtainable by moving t_i is at most the number of *value slots* actually hit by σ on the window. For T with $I(T) = \{\iota_1 < \cdots < \iota_r\}$ let $V(T) = \sigma(\{t_j : j \notin I(T)\})$ and let $M_i(T)$ be the number of distinct slots relative to $V(T)$ (a slot of u is $\#\{v \in V : v < u\}$) hit by the values $\sigma(p)$, $p \in (t_{i-1}, t_{i+1})$.

Lemma 1'. For every stable rule $\mathcal{J}$,

$$\mathrm{pat}_k(\sigma) \le \sum_T \min\Big(1,\ \prod_{j=1}^{r(T)} \frac{\min(M_{\iota_j}(T) + j - 1,\ k)}{b_{\iota_j}(T) - 1}\Big).$$

Proof. As in Lemma 1, group witnesses into classes $\mathrm{ext}(\psi)$; the number of distinct patterns in a class is trivially at most $|\mathrm{ext}(\psi)| = \prod(b_i - 1)$. For the other bound, insert the I-values $v_j = \sigma(t'_{\iota_j})$ one at a time into the sorted list of V and let s_j be the slot of v_j relative to $V_j = V \cup \{v_1, \dots, v_{j-1}\}$; the sequence $(s_1, \dots, s_r)$ determines the pattern. A prefix $(s_1, \dots, s_{j-1})$ determines the V-slot c_l of each earlier v_l and their relative order within each V-slot; for a window value u in V-slot c one has $\mathrm{slot}(u \mid V_j) = c + \#\{l < j : c_l < c\} + \#\{l < j : c_l = c,\ v_l < u\}$, and the last term lies in $[0, n_c]$ with $n_c = \#\{l < j : c_l = c\}$. Hence, over *all* realizations of the prefix, s_j takes at most $\sum_{c \text{ hit}}(1 + n_c) \le M_{\iota_j} + j - 1$ values, and also at most $|V_j| + 1 \le k$. Multiply over j. ■

(The per-factor cap $b_{\iota_j} - 1$ would be false here: realizations of the same prefix with different values v_l send a fixed window position to different slots. We caught this in a first draft by enumerating every class on the $\mathrm{sp}(6) = 17$ permutation; the corrected bound has no violations on all permutations tested.)

Theorem C' (corrected and certified). For all sufficiently large k,

$$\mathrm{sp}(k) > 1.0073\, k^2/e^2.$$

Finite reduction. Fix integers $3 \le \beta \le M \le n - 2$, select even indices whose widths satisfy $b_i \ge \beta$, and let $W(T)$ be the summand of Lemma 1'. Let B be independent Bernoulli(q) sites on the positive integers, $q = 1 - x$, and $T(B)$ its first k sites. Put $W = 0$ when $t_k > n$; all value sets use $B \cap [n]$. Since $\Pr(T(B) = T) = q^k x^{t_k - k}$,

$$\mathrm{pat}_k(\sigma) \le A(n,k,x)\, \mathbb{E}W,$$
$$A(n,k,x) = x^{-n-1}(x/q)^{k+1}.$$

The $\ell = \lfloor (k-1)/2 \rfloor$ even widths are independent with law $P(b) = (b-1)q^2x^{b-2}$, $b \geq 2$.

For a fixed window of width $b \leq M$, its $b-1$ internal values determine $d_b = b-2$ gaps. The sets of external positions whose values fall in those gaps are disjoint, have total size at most $n-b-1$, and are independent of the sites inside the window. Their Bernoulli hit count S satisfies $M_i \leq S+3$. For $z \geq 0$, concavity and monotonicity of $a \mapsto \log(e^z - (e^z-1)x^a)$ give

$$\begin{aligned} \mathbb{E}e^{zS} &\leq (1 - p_b + p_b e^z)^{d_b}, \\ p_b &= 1 - x^{(n-b-1)/d_b}. \end{aligned}$$

For a fixed $R > 0$, choose the upper root u_b of $d_b D(u_b \| p_b) = Rk$, or set $u_b = 1$ if this root does not exist, and put $z_b = d_b u_b$. Then $\Pr(S > z_b) \leq e^{-Rk}$; at the cap the probability is zero. Crucially, these definitions apply only to $\beta \leq b \leq M$.

Let H be the event that every even width is at most M. On the good event the slot product is bounded using $M_i \leq z_b + 3$; elsewhere $W \leq 1$. A fixed window having both endpoints selected and exactly one internal site has probability $qP(b)$. Its external hit event is independent of this selection. A union bound over indices, positions and widths therefore gives

$$\begin{aligned} \mathbb{E}W \leq \mathbb{E}\left[\mathbf{1}_H \prod_{j=1}^{r} \frac{z_{b_{i_j}} + j + 2}{b_{i_j} - 1} \right] \\ + \ell n q e^{-Rk} + \ell \Pr(b > M). \end{aligned} \tag{C'}$$

The original unbounded-width formula lacked the last term and evaluated invalid slot means beyond the host. Equation (C') avoids that defect.

Uniform minimum and limiting rate. Fix the exact rational parameters

$$\begin{aligned} \lambda = 1.0073, \quad \tau = 0.96813, \quad \beta_0 = 0.53692, \\ R = 0.00783, \quad c_0 = 0.4805, \quad \theta = \tau e^2/\lambda, \end{aligned}$$

and take $n = \lfloor \lambda k^2/e^2 \rfloor$, $x = e^{-\theta/k}$, $\beta = \lceil \beta_0 k \rceil$, $M = \lfloor n/4 \rfloor$. For $t \geq \beta_0$ let $p(t) = 1 - e^{-\tau/t}$ and let $u(t) > p(t)$ solve $tD(u(t)\|p(t)) = R$. This root is below one: it suffices to check the inequality at β_0, since $t\log(1/p(t))$ increases.

The function $u(t)$ is nonincreasing: for fixed $u \geq p(t_1)$, both t and $D(u\|p(t))$ increase for $t \geq t_1$. Also $c(t) = tu(t)$ is nondecreasing. To prove this, set

$$\Lambda_t(z) = t\log(1 + (e^z - 1)(1 - e^{-\tau/t})).$$

Writing this as $tg(\tau/t)$, the function g is concave with $g(0) = 0$, hence $\Lambda_t(z)$ increases in t. Its upper-tail Legendre transform decreases in t for each fixed argument; the upper level-R endpoint $c(t)$ consequently increases.

The interval certificate checks $c_0 < \beta_0 u(\beta_0)$ and $c_0 < 1 - e^{-3\tau/4}$. Uniform convergence of the finite roots on $\beta \leq b \leq k$ then gives $z_b \geq c_0 k$ there for large k. For $k \leq b \leq n/4$, use $z_b \geq d_b p_b$ and the increasing function $t(1 - e^{-a/t})$ to obtain the same bound. This proves the needed minimum analytically, without a grid search over widths.

Put $g_k(b) = z_b/(b-1) \leq 1$. Since $\sum_{j \leq r}(j+2) = (r^2+5r)/2$, the first term of (C') is at most

$$\mathbb{E}\left[\mathbf{1}_H \prod_{\text{selected}} g_k(b) \exp\left(\frac{r^2+5r}{2c_0 k} \right) \right].$$

An exact Gaussian integral for the quadratic exponential and independence of the widths show that its upper exponential rate is at most $\sup_{s\in\mathbb{R}} F(s)$, where

$$\begin{aligned} F(s) &= -c_0 s^2/2 + \tfrac{1}{2}\log(P_0 + Ge^s), \\ P_0 &= 1 - (1+\theta\beta_0)e^{-\theta\beta_0}, \\ G &= \int_{\beta_0}^{\infty} \theta^2 t e^{-\theta t} u(t)\, dt. \end{aligned}$$

Indeed the finite integral has bracket $[P_k + e^{s+5/(2c_0k)}G_k]^{\ell}$, with $P_k \to P_0$, $G_k \to G$. Bounded weights and tightness give convergence on compact s intervals; a negative quadratic plus a linear term controls the tails. Furthermore $F'' \le -c_0 + 1/8 < 0$, so at any s_0,

$$\sup_s F(s) \le F(s_0) + \frac{F'(s_0)^2}{2(c_0 - 1/8)}.$$

This controls the entire real line, including between numerical evaluation points.

The exact width tail is $\Pr(b > M) = x^{M-1}[1 + (M-1)q]$, with rate $-\tau/4$. Stirling's formula gives the prefactor rate $P = \tau - 1 - \log\tau + \log\lambda$. Thus

$$\limsup_{k\to\infty} \frac{1}{k}\log\frac{\mathrm{pat}_k(\sigma)}{k!} \le P + \max\{\sup_s F(s), -R, -\tau/4\},$$

uniformly in σ.

Certificate. The program in W25 uses outward Decimal intervals, 8192 Darboux panels on $[\beta_0, 4]$, the monotonicity of u, exact Gamma interval masses, and a Gamma tail bound beyond 4. Suggested root brackets are accepted only after interval relative-entropy sign checks. It obtains

$$0.08638273335 < G < 0.08640436421.$$

At $s_0 = 0.1005$, the certified upper bounds for the good, bad-window and width-tail rates, including P, are respectively less than

$$-0.00003104, \qquad -0.00003761, \qquad -0.23424011.$$

All are strictly negative. Hence every host of length $\lfloor 1.0073k^2/e^2 \rfloor$ misses a pattern for all large k, proving the theorem. The full finite proof and exact enclosures are in W25; this is an interval certificate, not a Lean formalization. □

3 Short superpatterns

Theorem 2. $\mathrm{sp}(7) \le 23$ and $\mathrm{sp}(8) \le 30$. Witnesses are

$\sigma_7 = 7\ 20\ 13\ 10\ 2\ 18\ 23\ 4\ 12\ 16\ 8\ 5\ 19\ 15\ 1\ 9\ 22\ 14\ 6\ 17\ 11\ 3\ 21$,

$\sigma_8 = 13\ 4\ 25\ 18\ 8\ 30\ 12\ 22\ 1\ 28\ 19\ 10\ 6\ 14\ 24\ 27\ 3\ 16\ 21\ 11\ 5\ 20\ 29\ 15\ 7\ 23\ 2\ 17\ 26\ 9$.

Verification. Exhaustive subset standardization finds all 5040 and 40320 patterns, respectively. The Python checker examines 245157 and 5852925 subsets; a separate C checker uses a different enumeration. The Lean project proves soundness of its DFS checker and verifies these instances

using `native_decide`, whose compiler trust is stated in §8. Simulated-annealing logs and additional witnesses are archived in W1 and `experiments/witnesses/`.

No completed nonexistence proof at length 22 was found in W6. The elementary implication $\mathrm{sp}(k+1) \geq \mathrm{sp}(k)+1$, together with $\mathrm{sp}(6) = 17$ [2], gives only $18 \leq \mathrm{sp}(7) \leq 23$ and $19 \leq \mathrm{sp}(8) \leq 30$ here. To see the implication, delete the maximum of a $(k+1)$-superpattern: each k-pattern occurs as the first block of a copy of $\pi \oplus 1$, whose first k entries cannot contain that maximum. The former draft's lower bounds 19 and 20 did not follow from this argument.

4 Random targets and a certified lookahead bound

Write Π_N for a Poisson process of intensity N on the unit square, so its expected number of points is N. By adjoining points from an independent iid sequence, a containment result at Π_{Ck^2} transfers to a uniform host of length $\lceil (C+\varepsilon)k^2 \rceil$, at cost $e^{-\Omega_\varepsilon(k^2)}$. All constants below use this normalization.

Theorem 17. For every fixed $C > 0.4649$, a uniform target $\pi \in S_k$ is contained in an independent uniform permutation of length $\lceil Ck^2 \rceil$ with probability $1-o(1)$. Equivalently, all but an $o(1)$ fraction of targets have individual failure probability $o(1)$.

Proof outline and certificate. Fix a strip size h. Partition target values into consecutive groups of h, and partition the host into corresponding horizontal strips. Scaled coordinates have clock budget k, strip height h and intensity C. In the gap (y_L, y_R) of the current target value, let m_b, m_a be the numbers of unplaced values below and above it. Search beyond a safe clock $a_s = \max(a, e_j)$, where e_j bounds every previous exploration in the strip, and choose the point minimizing

$$(x - a_s) + \frac{1}{C}\left(\frac{\bar{V}_{m_b}}{y - y_L} + \frac{\bar{V}_{m_a}}{y_R - y} \right).$$

Restrict each side that still holds target values by a fixed relative margin ε. Every unfinished gap then has height at least $\varepsilon^{h-1}h$. The sublevel set explored by this minimization lies to the left of a finite safe edge, so each subsequent fresh search has the required conditional Poisson law.

Conditional one-step laws depend only on the strip's own state. Thus the fresh costs of the strips are independent, and for a uniform target they are identically distributed with finite exponential moments. Their mean is $\mathcal{V}_h/(Ch)$. A safe-clock collision costs at most a constant D_h, and needs fewer than a fixed distance of clock advance between two visits. For n intermediate visits its probability is at most $(eChD_h/n)^n$. Taking $J = \max(\lceil e^2 ChD_h \rceil, \lceil 3 \log k \rceil)$, the expected number of visits with fewer than J intermediate visits in the same strip is at most Jh. Here the position distance ranges from 1 through J, inclusive. The expected collision cost is $O_h(\log k)$, and Chernoff plus Markov gives failure $e^{-\eta k} + O_h(\log k/k)$ whenever $C > \mathcal{V}_h/h^2$.

For a unit-height gap at intensity one, define

$$W_\varepsilon(A,B) = \int_0^\infty e^{-a_{A,B}(s)}\, ds,$$
$$a_{A,B}(s) = \int_{I_\varepsilon} \left(s - \frac{A}{y} - \frac{B}{1-y} \right)_+ dy,$$

where a zero potential omits its term and removes that side's margin. Monotonicity and induction

show $\mathcal{V}_n \le \bar{V}_n$ if

$$\bar{V}_0 = 0, \qquad \frac{1}{n}\sum_{m=0}^{n-1} W_\varepsilon(\bar{V}_m, \bar{V}_{n-1-m}) \le \bar{V}_n.$$

The independent MPFR checker verifies all 512 inequalities for the exact dyadic margin $\varepsilon = 1/64$, treating the stored potentials as exact dyadic candidates. It gives

$$\bar{V}_{512}/512^2 \le 0.46487433620981994 < 0.4649.$$

The old floating quadrature is not trusted in this verification. Directed root and logarithm evaluations enclose a and its derivative; convex tangent integration and an exponential tail give upper bounds for W. Every intermediate recurrence and its positive margin is saved.

For arbitrary k, pad to $k' = h\lceil k/h \rceil$. The first k entries of a uniform target in $S_{k'}$ standardize to a uniform target in S_k. A fixed coefficient margin absorbs $k' - k = O_h(1)$ and de-Poissonization. Finally Markov's inequality over targets gives the stated individual-target interpretation. Full details and compilation commands are in W31. □

The constants hidden in $O_h(\log k/k)$ are very large for $h = 512$; this is an asymptotic theorem without a practical finite-k bound. The unrestricted Bellman optimum near 0.4623 is a numerical extrapolation. Its expected-cost optimality proof requires future target ranks to be unobserved. W32 extends a corresponding expected-cost inequality under explicit freshness and nonanticipation hypotheses; high-probability failure additionally requires a separately proved weak law. These statements do not cover retained candidates or general global embedding algorithms.

5 Structured classes at Alon's coefficient

5.1 Independent-strip cost and the repaired grid reduction

Using zero-based positions and values, a block-grid pattern has $k = rh$ and

$$\pi_\tau(ir + s) = \tau_i(s)h + i, \qquad 0 \le i < h, \quad 0 \le s < r,$$

where each τ_i permutes $\{0, \dots, r-1\}$. It visits one point in each of r value strips per row, increasing within each strip across rows. A tilted grid has every τ_i equal to the identity. Assume the stronger revisit condition H_η: successive visits to a strip have index difference at least ηr, for a fixed $\eta > 0$.

For independent unit Poisson quadrants $\Psi_1, \dots, \Psi_b$, define

$$K_b = \min\left\{ x_b + \sum_{j=1}^{b} y_j : (x_j, y_j) \in \Psi_j,\ 0 < x_1 < \cdots < x_b \right\},$$

$$C_b^{\mathrm{mix}} = \left(\frac{\mathbb{E}K_b + \mathbb{E}K_{b+1}}{2(2b+1)} \right)^2.$$

Theorem 18 (repaired). Fix $b \ge 1$, $\eta, A > 0$ and $C > C_b^{\mathrm{mix}}$. Uniformly over individual block-grid patterns satisfying H_η and $\min(r, h) \ge \log^3 k$, the probability of failure in the fixed-strip model of Π_{Ck^2} is at most k^{-A} for all sufficiently large k.

Proof. Use a stationary renewal partition of visits into lengths $b, b+1$. At each block inspect fresh squares of side L beyond a safe clock, and choose the ordered path minimizing total x advance plus

y increments. On an infeasible block, mark failure and continue with artificial increments L; this defines all variables without conditioning on success. Given the partition, the relative block outputs are independent and bounded by L. Their means converge as $L \to \infty$ to the unwindowed means, since the optimal cost has a Gaussian tail bound from a sequence of corner searches. Scaling and differentiation of the weighted cost give equal unwindowed mean x cost and total y cost, namely $\mathbb{E}K_b/(2\sqrt{C})$.

Take $L^2 = (b+1)(A+3)\log k/C$. Dividing the block's squares into successive subwindows shows that the probability of any infeasible block is at most $(b+1)k^{-A-2}$. Between revisits there are $m \geq \eta r/(2(b+1))$ complete blocks for large k. Each advances the clock by at least an independent $\min(L, E)$ with $E \sim \mathrm{Exp}(CL)$. Thus the probability of any safe-clock offset is at most $k(eCL^2/m)^m$, smaller than every inverse power of k.

The two stationary mean rewards per visit, ρ_x^L and ρ_y^L, both converge to

$$\rho = \frac{\mathbb{E}K_b + \mathbb{E}K_{b+1}}{2\sqrt{C}(2b+1)} < 1.$$

They need not be equal at finite L. For the x budget, apply Hoeffding to the centered iid full-block rewards $\mathbb{E}X_D^L - \rho_x^L D$, taking a union bound over possible numbers of blocks. This controls the random conditional mean by $\rho_x^L k + o(k)$. Conditional Hoeffding then controls the actual total x cost, with error $e^{-\Omega(k/L^2)}$.

For a fixed strip, the renewal phase is a finite irreducible aperiodic chain. Its states at visits separated by ηr couple to independent stationary phases with error at most $hB_b e^{-c_b \eta r}$. Hoeffding first controls the sum of the phase-dependent conditional y means and then, conditionally on the partition, the actual y increments. The errors are $e^{-\Omega(h)} + hB_b e^{-c_b \eta r} + e^{-\Omega(h/L^2)}$. The initial and final partial blocks each cost at most L. Union over r strips; $r, h \geq \log^3 k$ makes all budget and offset errors smaller than every inverse power. On the remaining event every chosen point is real and all budgets fit, giving the copy. W34's `reduction.md` supplies the complete construction and bounds. □

Theorem 20 (independent-strip constant). For all $n \geq 1$,

$$n \leq \mathbb{E}K_n \leq n + \sqrt{2n} + \tfrac{1}{2}.$$

Consequently $\gamma_\infty = \lim_n \mathbb{E}K_n/n = 1$ and $C_b^{\mathrm{mix}} \to 1/4$.

Proof. For $\rho > 0$, take a two-sided compound Poisson boundary G^ρ, normalized at zero, with jump rate ρ and independent exponential jumps of mean ρ. For an independent Poisson quadrant set

$$T_\Psi B(a) = \min_{x_p > a} \{y_p + B(x_p)\}.$$

Its centered output increments have the same compound Poisson law, while $T_\Psi G^\rho(0) - G^\rho(0)$ is exponential of mean ρ. Here stationarity concerns centered increments, not the intercept.

For completeness, the gap $D(a) = T_\Psi G^\rho(a) - G^\rho(a)$ has survival function solving

$$\varphi(z) = \frac{\rho}{\rho + z}\left(e^{-z/\rho} + \int_0^z \rho^{-1} e^{-j/\rho}\varphi(z-j)\,dj\right).$$

The solution is $e^{-z/\rho}$, with uniqueness by the integral inequality for the difference of two solutions. Reading from right to left, D jumps upward at rate ρ by an exponential amount of mean ρ, and

downward at rate D to a uniform point below its current value. The exponential density satisfies detailed balance. Time reversal exchanges the boundary jumps with the output jumps, proving the centered stationarity law. The full marked-process argument is in W36.

The operator is monotone and commutes with constants. Compare G^ρ with the linear boundary x using the one-sided suprema of $x - G^\rho(x)$ and its negative. The exponential martingale bounds give

$$\mathbb{E}K_n \le n\rho + \rho/(\rho^2 - 1), \quad \rho > 1,$$
$$\mathbb{E}K_n \ge n\rho - \rho/(1 - \rho^2), \quad 0 < \rho < 1.$$

Taking $\rho = 1 \pm 1/\sqrt{2n}$ yields the two-sided estimate $n - \sqrt{2n} \le \mathbb{E}K_n \le n + \sqrt{2n} + 1/2$, hence the limit. Concatenation across independent later strips gives subadditivity of $\mathbb{E}K_n$; Fekete's lemma and the limit then strengthen the lower bound to $\mathbb{E}K_n \ge n$ for every n. □

Corollary (simultaneous tilted grids). For every fixed $\varepsilon > 0$, a uniform host of length $\lceil (1/4 + \varepsilon)k^2 \rceil$ simultaneously contains every tilted grid with $rh = k$, $\min(r, h) \ge \log^3 k$, and every dihedral image of each, with failure $O(k^{-2}) + e^{-\Omega_\varepsilon(k^2)}$.

Proof. Choose a fixed b with $C_b^{\mathrm{mix}} < 1/4 + \varepsilon/2$. Tilted grids satisfy H_1. Apply Theorem 18 with $A = 3$ and union bound over at most $8k$ shapes and images, then de-Poissonize. □

For the larger H_η class, Theorem 18 remains an individual-pattern statement; its cardinality can be much larger than polynomial. Also $\gamma_\infty = 1$ is a statement about the first-passage functional. It gives no matching lower containment threshold for tilted grids.

5.2 One square event for every monotone inflation

An inflation $\rho[\alpha_1, \dots, \alpha_m]$ replaces each entry of $\rho \in S_m$ by a block occupying consecutive positions and consecutive values, with the relative value order of the blocks prescribed by ρ.

Theorem 22 (shared squares). Fix $\varepsilon, A > 0$. There is $K = K_{\varepsilon,A}$ such that a uniform permutation of length $\lceil (1/4 + \varepsilon)k^2 \rceil$ simultaneously contains all inflations of arbitrary skeletons by monotone blocks of total length k, each of length at least $\lceil K\sqrt{\log k} \rceil$, with failure at most k^{-A} for all large k. Block sizes may differ and their directions may be chosen independently.

Proof. For every $C > 1/4$, the lower-tail theorem of Deuschel and Zeitouni [5, Thm 1] implies that a square with Poisson mean Ca^2 lacks an increasing subsequence of length a with probability at most $e^{-c_C a^2}$ for all sufficiently large integers a. To obtain this precise Poisson version, choose $1/4 < C_1 < C$ and $1/\sqrt{C_1} < x < 2$, retain $\lfloor C_1 a^2 \rfloor$ points, and combine the Poisson count tail with their fixed-size lower-tail theorem. Reflection gives the decreasing version. No uniform near-edge asymptotic is used.

Work at intensity Ck^2, $C = 1/4 + \varepsilon/2$. For integers a, s, t with $L \le a \le k$ and $0 \le s, t \le k - a$, consider

$$Q(s, t, a) = (s/k, (s + a)/k) \times (t/k, (t + a)/k).$$

There are at most $(k+1)^3$ such squares. With probability at least $1 - 2(k+1)^3 e^{-c_C L^2}$, each contains both required monotone subsequences of length a.

Fix any skeleton ρ and sizes $a_1, \dots, a_m \ge L$ summing to k. Put

$$s_i = \sum_{j<i} a_j, \qquad t_i = \sum_{\rho(j)<\rho(i)} a_j.$$

Choose the required monotone block inside $Q(s_i, t_i, a_i)$. The x intervals are ordered by i and the disjoint y intervals by $\rho(i)$, so these points give the inflation. The event was defined without reference to the skeleton, sizes or directions, proving simultaneous containment. Set $L = \lceil K\sqrt{\log k} \rceil$ with $c_C K^2 > A + 4$, and de-Poissonize. □

For direct and skew sums alone, $O(k^2)$ squares suffice. This strengthens the defect-free subclass of the older W37 box argument. It does not cover general interleavings, short defect blocks, or repeated 21 with blocks of length two. Its priority as a standalone corollary has not been established.

6 Exact frontier and bounded diagnostics

6.1 Repeated 21

Let L_{21} be the maximum number of direct-summed decreasing pairs in a host and $c_{21} = \lim L_{21}(N)/\sqrt{N}$. Alon's conjecture requires $c_{21} \geq 1$. Renewal rules give $c_{21} \geq 0.7866$; §6.2 proves $c_{21} \leq 1$ and reports numerical evidence that $c_{21} < 1$. To go beyond local renewal rules, W40 records all feasible partial copies.

Proposition 23 (exact global update). Scan points in increasing x order. Let F_m be the least possible maximum y coordinate of a completed m-pair copy, with $F_0 = 0$ and an absent state assigned infinity. For every previous apex z, record the interval $(F_m^{\text{before } z}, z)$ and define

$$H_m(y) = \min\{z : F_m^{\text{before } z} < y < z\}.$$

On arrival at height y, using the old state, update

$$\begin{aligned} F_{m+1} &\leftarrow \min(F_{m+1}, H_m(y)), \\ H_m(t) &\leftarrow \min(H_m(t), y) \qquad \text{for } F_m^{\text{old}} < t < y. \end{aligned}$$

Proof. A newly completed copy ends at y, with an earlier apex z and a completed m-pair copy before z whose maximum lies below y. This is exactly a recorded interval covering y; conversely every such interval supplies a witness. A new pending pair starts at y above an old completed copy, giving the second update. Using the old state prevents reuse of the arriving point. □

The original implementation used one segment tree per level and required $O(n(M+1)\log n)$ time and $O(n(M+1))$ space, with $M = L_{21}$. The following dominance argument removes that factor of $M + 1$.

Proposition 24 (permanent pruning and localization). Retain a level-m pending interval (l, z) only while $z < F_{m+1}$. On arrival of a fresh height y, let $F_j < y < b = F_{j+1}$. Among all retained intervals find the least apex z whose interval covers y. If it exists, replace F_{j+1} by z and delete all retained apices in $[z, b)$. Finally insert the single interval (F_j, y). This preserves every completed threshold and admits an $O(n \log n)$-time, $O(n)$-space implementation.

Proof. The finite thresholds strictly increase with m: deleting the last pair from a minimizing copy leaves a copy with smaller maximum. Each threshold decreases with time. Thus an interval with $z \geq F_{m+1}$ can never improve that threshold, now or later, and any future extension from it is dominated by the already completed copy. It can be discarded permanently.

Every retained interval consequently satisfies $F_m \leq l < z < F_{m+1}$. At height y, levels below j have all apices below y, and levels above j have all activation marks above y. Only level j can improve;

its chosen apex obeys $y < z < b$. Deleting $[z, b)$ removes exactly its newly dominated intervals. A new interval at a level below j would be immediately dominated, and higher levels cannot start at y. Only (F_j, y) survives. This uses the old F_j, since only F_{j+1} changes, proving the invariant by induction.

The current gap of an apex identifies its level. Store all intervals in one segment tree, indexed by apex and carrying the minimum lower mark in each subtree, with lazy range deletion. Binary search finds j; a first-cover query, range deletion and insertion each take $O(\log n)$ time. The tree and threshold array use $O(n)$ space. □

The implementation matches an independent unpruned recurrence on every completed threshold of 372,249 prefix states: all 46,233 hosts of lengths at most 8, 100 random hosts and nine extreme hosts. Final answers on the exhaustive hosts also agree with brute-force subsequences. At $n = 4096$, the earlier 64 seeded hosts give mean $L_{21}/\sqrt{n} = 0.9411621$, standard error 0.0031655. This finite-host statistic gives no asymptotic bound.

For prior art, Albert et al. give $O(n^2)$ for longest subsequences with layers of sizes one or two in section 3.7, p. 236 [8]; Albert later reports $O(n \log n)$ for this class in section 4 of the 2005 preprint [9]. Their objective is total length, whereas ours counts complete two-point layers, equivalently a weighted score on that class. A weighted adaptation may already give this complexity. We make no algorithmic priority claim; the explicit marked state is the tool for the present probabilistic investigation.

Pruning does not justify erasing activation marks. Prefixes $(3, 2, 4, 1)$ and $(2, 3, 1, 4)$ have the same thresholds $(0, 2, \infty)$ and the same retained apices $\{1, 4\}$. Their apex-4 intervals are respectively $(3, 4)$ and $(2, 4)$, so the next height $5/2$ completes a second pair only in the latter state.

There is an exact flux identity for the marked state S_t under unit-intensity Poisson arrivals in a finite-height strip. For a fixed cut u, let $N_u = \#\{m \geq 1 : F_m \leq u\}$ and $j = N_u$. Then

$$r_u(S) = \left| \bigcup_{(l,z):\, F_j < z \leq u} (l, z) \right|,$$

$$\mathbb{E}N_u(t) = \int_0^t \mathbb{E}r_u(S_s)\, ds$$

for the process started empty, where the union uses retained intervals. Indeed, N_u increases precisely when the arriving height is covered by such an interval; the least covering apex also lies below u. Each increase is by one, and Poisson intensity equals interval length. This gives a concrete condition for a stationary-law calculation, but does not itself establish one or identify c_{21}.

6.2 Layered patterns are at least as hard as the identity

Throughout this subsection a configuration P is a finite planar point set with distinct abscissae and distinct ordinates, $z < w$ means strict inequality in both coordinates, and for a point a north-west of b we write $a \wedge b = (x_a, y_b)$ and $a \vee b = (x_b, y_a)$. Fix patterns α, β, possibly empty, and put

$$U = \{z : \text{ the points of } P \text{ below-left of } z \text{ contain } \alpha\},$$
$$S = \{z : \text{ the points of } P \text{ above-right of } z \text{ contain } \beta\}.$$

U is an up-set and S a down-set. The *inside* points are $I = P \cap U \cap S$, the others are *outside*; write $O = P \setminus I$, and X, Y for the coordinate sets of I. Reading the middle block of a copy shows that P

contains $\alpha \oplus 12 \oplus \beta$ exactly when two inside points form an increasing pair, and contains $\alpha \oplus 21 \oplus \beta$ exactly when two inside points a, b, with a north-west of b, satisfy $a \wedge b \in U$ and $a \vee b \in S$; call such a pair a *D-inversion*.

Theorem 26 (block flip). For all patterns α, β and every n, the number of $\sigma \in S_n$ containing $\alpha \oplus 21 \oplus \beta$ is at most the number containing $\alpha \oplus 12 \oplus \beta$. More precisely, partition S_n into fibres according to (O, X, Y). Each fibre with $I \neq \emptyset$ consists of all bijections $f : X \to Y$ whose graph lies in $U \cap S$ (the *valid fillings*), and the difference of the two counts equals $\sum(a-1)$ over fibres, where $a \geq 1$ is the number of valid fillings without a D-inversion.

Proof. First, U and S are determined by the outside points. Indeed, if $z \in U$, choose among the copies of α below-left of z one whose rightmost point is leftmost possible; if it used an inside point p, a copy of α below-left of p would contradict the choice. The same argument, applied to a configuration $O \cup \Gamma_f$ built from a valid filling f, shows that its outside points are again exactly O; so a fibre is the set of valid fillings and U, S are constant on it. Second, two sorting arguments. If a valid filling has an increasing pair $p < q$, replace it by $(x_p, y_q), (x_q, y_p)$: both lie between $p \in U$ and $q \in S$, so the filling stays valid and has more inversions; hence the decreasing bijection is valid and is the only valid filling without an increasing pair. If a valid filling has a D-inversion (a, b), replace it by $a \wedge b, a \vee b$: now $a \wedge b \in U$ by assumption and $a \wedge b \leq a \in S$, while $a \vee b \in S$ by assumption and $a \vee b \geq a \in U$; the filling stays valid and has fewer inversions, so some valid filling has no D-inversion. A fibre with F fillings therefore has $F - 1$ members containing $\alpha \oplus 12 \oplus \beta$ and $F - a$ members containing $\alpha \oplus 21 \oplus \beta$, with $a \geq 1$. The details are written out in the W90 file `proof.md`. ■

Corollary 27. For all n and m, a uniform σ_n satisfies

$$\Pr(\sigma_n \supseteq 21^{\oplus m}) \leq \Pr(\mathrm{LIS}(\sigma_n) \geq 2m).$$

Hence $L_{21}(\sigma_n)$ is stochastically dominated by $\lfloor \mathrm{LIS}(\sigma_n)/2 \rfloor$ and $c_{21} \leq 1$. More generally, flipping blocks 12 to 21 in a direct sum never increases the containment probability.

Proof. Apply Theorem 26 m times, with $\alpha = 12^{\oplus(m-1-i)}$ and $\beta = 21^{\oplus i}$ for $i = 0, \ldots, m-1$, and use $\mathbb{E}\,\mathrm{LIS}(\sigma_n) \sim 2\sqrt{n}$ [10, 11]. ■

Proposition 28 (equality and the smallest defect). If α or β is empty then $a = 1$ in every fibre, so the two counts in Theorem 26 are equal; this is the case $k = 2$ of [12]. A fibre with at most two inside points has $a = 1$. A fibre with three inside points has $a = 2$ exactly when its board is the full 3×3 grid $X \times Y$ minus the lowest-left cell, which is not in U, and the highest-right cell, which is not in S; then the fillings are 321, 231, 312 and the last two have no D-inversion.

The first assertion follows by induction on the top row: in a filling without D-inversions its point must occupy the last available cell of that row. The others are a finite enumeration. In the three-point case the inversion pair straddles an outer corner of the explored region: each of its points can start an increasing continuation, but the pair cannot serve as the next block 21. The smallest instances are the fibre of 3416725 for 1324 against 1234, as in [7], and a single fibre at $n = 9$ for 214365 against 123456. Exhaustive enumeration confirms $|\mathrm{Av}_n(214365)| = |\mathrm{Av}_n(123456)|$ for $n \leq 8$ and a difference of exactly one at $n = 9$.

Theorem 26 is the transversal case $k = 2$ of the theorem of Jelínek and Karpilovskij on fillings of skew shapes [6], who conjecture the analogue for chains of every size and note that it is known only for $k \leq 2$. The short proof above is included for completeness.

The deficit. By Corollary 27, $D(n) = 2(\mathbb{E}\lfloor \mathrm{LIS}(\sigma_n)/2 \rfloor - \mathbb{E}L_{21}(\sigma_n)) \geq 0$, and

$$c_{21} = 1 - \lim_{n\to\infty} \frac{D(n)}{2\sqrt{n}}.$$

Theorem 26 expresses $D(n)$ as a sum, over the m flips and over m, of the non-negative quantities $\mathbb{E}[(a-1)/F]$. We computed L_{21} and LIS on the same uniform permutation.

Table 2: Paired simulation of $D(n)$. Errors are one standard error.

n	samples	$D(n)$	$D(n)/\sqrt{n}$
10^3	700000	0.0736 ± 0.0040	0.00233 ± 0.00013
10^4	420000	0.2412 ± 0.0079	0.00241 ± 0.00008
10^5	140000	0.688 ± 0.021	0.00218 ± 0.00007
10^6	28000	2.304 ± 0.070	0.00230 ± 0.00007
10^7	2800	6.88 ± 0.34	0.00218 ± 0.00011
10^8	150	20.96 ± 2.18	0.00210 ± 0.00022

The weighted mean of $D(n)/\sqrt{n}$ is 0.002269 ± 0.000036, with $\chi^2 = 7.1$ on five degrees of freedom, and a power law fitted to the six points has exponent 0.491 ± 0.006. An independent program and random generator reproduce the values at $n = 10^4, 10^5, 10^6$. Numerically, therefore, $c_{21} \approx 0.99887$, and the pattern $21^{\oplus k/2}$ is contained with high probability only for n above about $1.0023\,k^2/4$. Converting only the first i blocks costs about $0.0025\,i$ in expected length, linearly in i. Patterns with decreasing layers of size $3, 4, 5, 6$ show $D(n)/\sqrt{n}$ near 0.0050, 0.0075, 0.0095, 0.0105, and layers of size 8 to 16 between 0.012 and 0.017; for layers of size at least three the inequality itself is the conjecture of [6].

Status. The statement $c_{21} < 1$, which would refute Alon's conjecture at the coefficient $1/4$, is supported by these data but is not proved. The increasing permutation is known not to be extremal for online embedding [4], and the present evidence indicates that it is not extremal offline either. A proof needs a lower bound of order $\sqrt{n}$ for the sum of the terms $\mathbb{E}[(a-1)/F]$. The repository's W90 notes reduce this to a statement about where the longest increasing chain beyond an explored front can start; that statement is open.

6.3 Canonical second moments

Let Y_π count copies of π with no single-point left move preserving the pattern. Such a copy exists if and only if an ordinary copy exists: minimize the sum of its position coordinates. For a uniform target independent of the host, write

$$\mu_{\mathrm{can}} = \mathbb{E}_\pi \mathbb{E} Y_\pi,$$
$$R_{\mathrm{can}} = \frac{\mathbb{E}_\pi \mathbb{E} Y_\pi^2}{\mu_{\mathrm{can}}^2} = \frac{1}{\mu_{\mathrm{can}}} + D.$$

Cauchy–Schwarz gives averaged containment probability at least $1/R_{\mathrm{can}}$. Distinct canonical copies of the same pattern cannot share $k-1$ points (W38, Lemma 3). This exact rigidity statement does not control all other overlaps.

Table 3: Bounded follow-up at actual $N/k^2 = 1/4$. Intervals are 95% paired bootstrap intervals over hosts, preserving covariance; they are not rigorous numerical enclosures.

k	Hosts	R_{can}	$1/\mu_{\mathrm{can}}$	D and interval
6	1000	20.21	19.22	0.98 [0.89, 1.08]
8	400	20.32	16.52	3.79 [3.53, 4.07]
10	64	20.35	13.40	6.94 [6.25, 7.71]

Every subset and every canonical overlap is enumerated within each sampled host; a separate Python implementation agrees on 35 small hosts. The flat total is partly a cancellation between a decreasing diagonal and an increasing off-diagonal part. The largest off-diagonal bin is $k-2$, but substantial mass remains at smaller overlaps. The four fixed-target diagnostics have broad overlapping intervals and do not establish pattern uniformity. All bins, including zeros, and all per-host data are retained in W38.

The plain-copy exponential rate and the proposed critical coefficient near $1/2$ remain heuristic. W12's canonical first-moment upper bound ceasing to decay near the numerical value 0.1925 cannot establish growth above that value; its transfer-operator supremum also lacks an interval certificate. Finally, superposing m independent intensity-N processes on the same unit square proves $p_\pi(mN) \le p_\pi(N)^m$ for avoidance probabilities. This does not turn a positive averaged success probability into typical-target high probability at the same coefficient.

6.4 Exact joint emptiness and surviving clusters

W41 resolves the finite joint-emptiness calculation. Fix overlap j and put $m = 2k - j$. A union type $T = (\rho, I, J)$ consists of $\rho \in S_m$ and two k-subsets covering $[m]$ with equal induced patterns. Retain it only if both copies are canonical within these m union points; this rejects cross-blocking by the other copy's exclusive points. Let $\mathcal{T}_{k,j}$ be these compatible types and let $Z_{k,j}$ count ordered canonical equal-pattern pairs, summed over targets.

Order the union's coordinates separately and write their $m+1$ gaps as $g, h \in \Delta_m$, where Δ_m has volume $1/m!$. Each canonical forbidden rectangle has endpoints among the union coordinates and the boundary. Set $D_{iq} = 1$ exactly when the open gap cell (i, q) belongs to at least one rectangle from either copy. The forbidden union has area $V_T = g^T D h$. Put $a_i(h) = \sum_q (1 - D_{iq}) h_q$.

Proposition 25 (exact overlap formula). For Poisson intensity λ on the unit square and a fixed host size $n \ge m$, respectively,

$$\mathbb{E} Z_{k,j}(\Pi_\lambda) = \lambda^m \sum_{T \in \mathcal{T}_{k,j}} \int_{\Delta_m^2} e^{-\lambda V_T}\, dg\, dh,$$

$$\begin{aligned} \mathbb{E} Z_{k,j}(\sigma_n) &= (n)_m \sum_{T \in \mathcal{T}_{k,j}} \int_{\Delta_m^2} (1 - V_T)^{n-m}\, dg\, dh \\ &= \sum_{T \in \mathcal{T}_{k,j}} \int_{\Delta_m} H_{n-m}(a_0(h), \dots, a_m(h))\, dh, \end{aligned}$$

where H_s is the complete homogeneous symmetric polynomial of degree s and $(n)_m = n!/(n-m)!$.

Proof. Represent a uniform host by iid square points. Each ordered pair has a unique x-ordered union and type. Choosing its labels contributes $\binom{n}{m}$ and ordering their x coordinates contributes

$m!$. Compatibility handles inserted points; all remaining points avoid the forbidden union with probability $(1 - V_T)^{n-m}$. This gives the fixed-host integral. Poisson-averaging it gives the first formula, equivalently the multivariate Mecke identity [13, Thm 4.4]. For $s = n - m$, expand $1 - V_T = \sum_i g_i a_i$ and use

$$\int_{\Delta_m} \prod_i g_i^{d_i}\, dg = \frac{\prod_i d_i!}{(s+m)!}, \qquad \sum_i d_i = s.$$

The multinomial factors cancel, leaving $s! H_s/(s+m)!$, whose prefactor cancels $(n)_m$. □

Writing $H_{n-m}(a(h)) = \sum_\alpha c_{T,\alpha} h^\alpha$, a second Dirichlet integration gives the nonnegative integer formula

$$n!\, \mathbb{E} Z_{k,j}(\sigma_n) = \sum_T \sum_\alpha c_{T,\alpha} \prod_q \alpha_q!.$$

For $j = k - 2$, all ten cases with $2 \le k \le 5$ and $k + 2 \le n \le 7$ agree exactly with independent exhaustive host enumeration. At $n = k + 2$ the ordered type counts are $8, 48, 276, 1648$. This is a finite formula; enumerating types is not efficient asymptotically, and no bounded second-moment ratio follows yet.

Two structural examples explain why local independence is insufficient. In the length-$(k + 2)$ permutation $2, 1, 4, 3, \ldots$ (with a final largest singleton when necessary), the first and last k positions form equal-pattern canonical copies. Their $k - 2$ common positions all change rank by two. Both copies are consecutive; the only possible preceding blockers for the second lie below its first selected value neighbour. Thus overlap $k - 2$ can shift the entire common correspondence.

More generally, in the deterministic host $21^{\oplus r}$, all copies of $21^{\oplus q}$ are canonical and correspond to choosing q whole blocks. Their number is $\binom{r}{q}$, and their overlap-$(2q - 2)$ ordered-pair count is $\binom{r}{q} q(r - q)$. Inversions occur only inside individual blocks, while skipped blocks lie below the next apex's lower neighbour, proving both claims. These examples obstruct a deterministic cluster-size argument; their frequency in random hosts still needs analysis.

6.5 Two-exchange selection: a finite obstruction to factorization

W42 retains a copy only if no lexicographically earlier same-pattern copy differs in at most two points. The first copy always survives, so existence is preserved. Two retained copies cannot intersect in $k-1$ or $k-2$ positions: the later would be rejected. Comparisons include earlier copies that were themselves rejected.

Its first moment is already target-dependent at $k = 3, n = 6$. Exact sums over all 720 hosts are $595, 598, 594, 594, 598, 595$ for targets $123, 132, 213, 231, 312, 321$, respectively. For comparison, the ordinary-canonical sum is 672 and the containment count is 588 for each target. Independent replacement-neighbourhood enumeration reproduces all six rows. Thus W12's pattern-independent first-moment factorization cannot be imported unchanged. The seven-cell census and verification code are in `experiments/w42-two-exchange/`.

The exact replacement formula in W42 conditions on the chosen copy: background points must avoid its one-point forbidden region and contain no pair that replaces two selected points to create an earlier copy. Integrating this no-pair event gives the first moment. Its target-dependent cost, including dependence among pairs sharing a point, is the next analytic problem. Finite averaged moment ratios improve in all seven cells, but no asymptotic ratio or containment threshold follows.

7 Corrections and preserved workstreams

The review changed both the accepted statements and the research queue.

- **Both-parity encoding withdrawn.** For $k = 5$, $T = (1, 2, 6, 9, 10)$ selects $I = \{3\}$, but changing only that position to obtain $T' = (1, 2, 3, 9, 10)$ changes the selected set to $\{4\}$. The proposed rule is not stable. Neither 1.000384 nor 1.000437 follows from it. The even rule and corrected C' are unaffected.
- **Slot checker corrected.** The per-class checker divided by the extension count a second time when summing classes. That invalidated its summed diagnostics, although the corrected per-class inequality still passed. The current checker fixes the sum and asserts the global bound as well.
- **Grid hypothesis strengthened.** No repetition within a fixed window of $b+1$ visits does not imply $\Omega(r)$ separation. For $r = 20, b = 2$, alternating the identity row with $(0, 1, 19, 2, \dots, 18)$ gives successive visits of strip 19 only three indices apart. Theorem 18 now uses H_η and controls conditional means rather than substituting stationary means.
- **Comparison-rate claim withdrawn.** The finite box inequality for $\pi^* = 1 \oplus \mathrm{dec}_{k-2} \oplus 1$ remains
$$p_{\pi^*}(N) \le p_{\mathrm{id}_k}(N(1-2\varepsilon)^2) + 2e^{-N\varepsilon^2}.$$
But $p_{\mathrm{id}_{k-2}} \le p_{\mathrm{id}_k}$, and a fixed-ratio large-deviation error $o(k^2)$ cannot imply an $o(k \log k)$ comparison uniformly near $N/k^2 = 1/4$. Those stronger inferences in W37 were unsupported.
- **Numerical limits and method restrictions.** W21's finite-size fits do not prove a typical-target limit near 0.22 or that identity is hardest among all patterns. W18 disproves a proposed lag lemma; it does not rule out every scheme using several threads. W22's concentration limitations concern its tested formulations.
- **September 2026 claims withdrawn.** A draft circulated in the repository between 22 and 27 September 2026 announced simultaneous universality at Ck^2 for all of S_k, the sharp coefficient $1/4$ for every class of bounded longest decreasing subsequence, $c_{21} = 1$, and finally Alon's conjecture. None of these is proved. The universality argument counted $e^{O(k)}$ certificates where a generic pattern needs $k!$, and required every one of about k^2 windows of bounded Poisson mean to be occupied, which fails at constant C. The bounded-LDS argument treated only skew sums of increasing blocks. The bound $c_{21} \le 1$ was obtained from the flux identity of §6 by a comparison whose sign was reversed; the bound is true, by Corollary 27, but $c_{21} = 1$ is contradicted by the data of §6.2. The corresponding workstreams W43–W89 are archived unedited with the audit.

The remaining workstreams are preserved in the full experiment index. W2 and W8 study structured construction obstructions. W9–W15, W18–W20, W23, W26 and W27 contain earlier thread, strip and cell-certificate approaches. W24, W28, W30 and W33 investigate missing-pattern counts, conditioning and probability slopes. These remain useful source material but are not all independently recertified by this review.

W4 reports small-alphabet equalities $f(k; k+1) = (k^2 + k)/2$ through $k = 5$, relating to Hunter's question [14], and explicit small rosaries related to [15]. The positive witnesses and the solver-reported nonexistence claims have different verification status: no independently checkable UNSAT proof artifact was identified for the latter. They should not be described as uniformly kernel-certified computational theorems. W6's length-22 search remains unresolved.

8 Verification and reproducibility

The repository's current result ledger gives commands, dependencies and scope. The main artifacts are:

- **C':** W25's `proof.md`, `certify_cprime.py` and `cprime_certificate.json`. Python Decimal outward arithmetic encloses the root brackets, integral and all three rates. Independent SciPy quadrature gives $G \approx 0.08639355273193$, strictly inside the interval; it is a cross-check, not the certificate.
- **Theorem 17:** W31's `certification.md`, `certify_mpfr.c` and the full TSV output. All 512 supersolution inequalities pass with exact margin $1/64$. Basic operations are widened outward; MPFR provides directed transcendental bounds. The quadratic discriminant is an exact dyadic polynomial evaluated at 256 bits before directed roots and division.
- **Witnesses:** `experiments/witnesses/` contains explicit permutations, exhaustive Python checking and recorded hashes. The Lean project additionally proves its checker sound and verifies the length-23 and length-30 instances.
- **Lean scope:** Hosted at https://github.com/adamhadani/superpatterns/tree/main/formal-verification/lean, the formalization includes the finite Theorem A, its rational-parameter negativity certificate at $\lambda = 1.0003$, Proposition 15's finite weighted inequalities, block-splitting combinatorics, the Erdős–Szekeres connection, and a sound superpattern checker with the length-23 and length-30 certificates. It contains no `sorry` and no added axioms; the certificate evaluations use `native_decide`. Nothing about random permutations, Poisson processes or asymptotic constants is formalized, and Theorem 26 is not formalized.
- **W38/W40/W41/W42:** separately written exhaustive checks verify the host enumerators, every threshold of the pruned frontier, and the exact joint-emptiness formula on small hosts. Exact cut-flux checks verify the marked-state identity. W42's independent replacement search checks 93,416 local decisions; an independent aggregate recount verifies its six-target first-moment counterexample.
- **W90 (§6.2):** `verify_fibers.py` recomputes U, S, the fibres and the boards from the definitions for seven choices of (α, β) and every $\sigma \in S_n$, $n \le 8$, and for $(21, 21)$ at $n = 9$; it reproduces $|\mathrm{Av}_8(1324)| - |\mathrm{Av}_8(1234)| = 26$. `exact_dist.c` enumerates S_n for $n \le 12$. The sweep computing L_{21} agrees with a cubic dynamic programme on 180,000 permutations and with the independently written W40 program on 81,240 permutations of length up to 10^6. All raw batch outputs are stored with the workstream.
- **Checkable certificates:** decimal certificate for C', directed MPFR for the 512 W31 inequalities, 372,249 prefix checks for W40, and the brute-force fibre identity for W90.

9 Acknowledgments and AI Assistance Disclosure

This is AI-assisted research. The author takes full personal responsibility for the mathematical claims, proofs, and specifications. The project used ChatGPT Codex for exploratory code and candidate recurrences, Claude Code for proofs, audits, numerical experiments and the results of §6.2, and, in September 2026, a many-agent harness [16]. The output of that September phase is the material withdrawn in §7. The history, counterexamples, scripts and raw outputs are preserved in the repository.

Open problems, in the order we would attack them: prove $c_{21} < 1$; prove the block-flip inequality for decreasing layers of size at least three; determine which pattern has the largest containment

threshold; and, independently of the constant, prove universality at Ck^2 for some absolute C, where the best bound remains $2000k^2 \log\log k$ [3]. The exact value of $\mathrm{sp}(7)$, which is 22 or 23, is also open.

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